

PHASE 2

Requirement Analysis Phase

AI StudyBuddy – AI-Powered Learning and Study Support Platform

Project Focus
AI-assisted learning resources, secure backend services, and personalized study support.

Phase Objective

To convert the initial idea into clear functional, technical, hardware, software, and user requirements for implementation.

2.1 Functional Requirements

- The system shall allow students to register and authenticate securely.
- Authenticated users shall be able to upload and manage study materials.
- The application shall generate summaries from supplied learning content.
- The application shall create AI-generated flashcards from study material.
- The application shall generate multiple-choice quizzes.
- The application shall prepare personalized study plans.
- Generated resources shall be retained for later access.
- Administrators shall be able to manage users and monitor system activity.

2.2 Security Requirements

- Protected APIs must verify valid JWT access tokens.
- Passwords must be stored using secure hashing rather than plain text.
- Role-based authorization must separate student and administrator capabilities.
- Sensitive configuration values such as database credentials and AI keys should be maintained through environment variables.

2.3 Software Requirements

- Windows 10/11, macOS, or Linux.
- Node.js version 16 or above.
- npm version 8 or above.
- Express.js for REST API development.
- MongoDB as the primary database.
- Postman for API verification.
- Visual Studio Code or another suitable development environment.

2.4 Hardware Requirements

- Intel Core i5 8th generation or newer, AMD Ryzen 5, or an equivalent processor.
- At least 8 GB RAM; 16 GB is preferable when several development services run together.
- At least 1 GB of available storage for the project workspace.

2.5 User Roles

- Student: registers, logs in, uploads learning material, requests AI-generated resources, and manages previously created resources.
- Administrator: manages accounts, observes platform activity, monitors AI usage, reviews logs, and supports system-level maintenance.

2.6 External Service Requirement

The backend requires access to the Google Gemini AI API for generating summaries, flashcards, quizzes, and personalized study plans. MongoDB connectivity is also required for persistent data storage.